

Hybrid Transactional/Analytical Processing (HTAP) Systems: Design and Performance Evaluation in Enterprise Databases

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Abstract: *HTAP systems which combine their Online Transaction Processing and Online Analytical Processing capabilities in one database system have become essential tools for achieving real-time data analysis. The research paper investigates an HTAP system which uses machine learning to enhance business performance through its operational technology. The proposed approach employs supervised learning models which include Random Forest and Gradient Boosting to identify workload patterns and forecast resource needs in real time. The model performs query optimization while distributing system resources and system load according to legal query patterns and system performance data. The research conducts experiments on TPC-H and Star Schema Benchmark standard benchmark datasets which contain mixed workload scenarios. The findings reveal that, in comparison to the conventional HTAP systems, the execution time of queries, the throughput and the latency are greatly improved. Moreover, the ML-based method is useful in diminishing the workload contention and increasing scalability. The research shows that the incorporation of machine learning in the HTAP systems can enhance the efficiency of the system to a long extent with faster and more effective real-time decision-making in enterprise database.*

Keywords: *HTAP, Machine Learning, OLTP, OLAP, Workload Prediction, Real-Time Analytics, Performance Optimization.*

I. INTRODUCTION

The introduction presets the background of the study by pointing out the increasing necessity to have real-time data processing in contemporary businesses. It states that in the conventional database systems, transactional (OLTP) and analytical (OLAP) works are separated which causes delays as a result of ETL processes as well as complexity of the system [1-3]. This builds a disconnection between data creation and data-driven decision-making.

A solution to this problem is then presented through the Hybrid Transactional Analytical Processing (HTAP) [4, 5]. HTAP systems are a hybrid of OLTP and OLAP that unite these two features into one architecture so that organizations are able to do real-time analytics on live transactional data.

The system will achieve faster operation through improved efficiency. The introduction demonstrates that HTAP systems function without any operational challenges according to the research findings. Running multiple workloads simultaneously results in three different problems which include workload interference, resource allocation problems, and difficulties in query optimization. The system expects to face operational challenges because of these difficulties which lead to performance degradation.

The introduction shows how machine learning solves its existing limitations through its dedicated purpose. The machine learning models predict three outcomes which include workload prediction and query classification and resource allocation optimization. The study establishes its main objective which involves creating and testing a machine learning-based HTAP framework that enhances database performance and database scalability and database real-time decision-making capabilities for enterprise databases.

A. Objectives of the Study

The study's main goal is to create a machine learning-based HTAP system which combines transaction processing with analytic processing into one system. The system will develop predictive models which classify workloads to enable dynamic resource allocation for system operational efficiency. The study evaluates system performance improvements through measurements of query execution time and throughput and latency. The research assesses the new method's performance in business environments by comparing it to existing HTAP systems.

B. Contribution of Study

Researchers developed an HTAP framework which uses machine learning to predict workload patterns while the system allocates resources to improve performance. The study shows that implementation of the actual system led to better execution time and throughput and latency performance

results. The study presents an adaptive method which decreases competition between workloads while it provides instant analytics for business databases.

B. Organization of the Paper

The paper is divided into a number of sections. The introduction provides the background and motivation of the study, and a literature review is given in detail with emphasis on all the existing research and identification of gaps. The methodology section describes the suggested machine learning-powered HTAP framework and experimental design. These findings and discussion part relate to performance evaluation using empirical data, discussion and comparative analysis. Lastly, the paper will have the conclusion of the paper with key findings, limitations and future research directions.

II. LITERATURE REVIEW

The literature review is based on the current studies on Hybrid Transactional Analytical Processing (HTAP) systems and usage of machine learning methods in database optimization [6-8]. The document begins by outlining traditional database system models which implement online transaction processing and online analytical processing as separate functions. Research from previous studies indicates that this separation leads to operational inefficiencies which include ETL processing delays and extended decision-making latency [9].

Recent research depicts HTAP systems as unified platforms which support both transactional operations and analytical processing. Researchers have examined multiple HTAP architecture types which include in-memory database systems and distributed architectural designs and hybrid storage models that merge row-based and column-based data storage methodologies [10, 11]. These reports prove that HTAP systems will be able to enhance real-time analytics and minimize the rate of data movement significantly.

The review also considers how machine learning can be used in database systems. The literature demonstrates that the ML methods are becoming popular to optimize queries, predict workloads, to select an indexing strategy, and to schedule resources [12-14]. Random Forest, Gradient Boosting and neural networks are some of the models that have been used effectively to predict query execution pattern and optimize system performance.

Nonetheless, limitations are also shown in the literature. The majority of HTAP systems are based on online techniques of optimization and do not include mechanisms of adapting to the dynamism of workloads. Also, there exists a lack of empirical studies to merge machine learning to HTAP architecture to improve performance. Aspects like contention of workloads, challenges of scalability and poor distribution of resources are not fully tackled.

Therefore, the literature review indicates that there is a definite gap in research: there is a lack of a machine learning-based framework of HTAP capable of learning how to manage the workloads of a system and enhance its performance. The study is based on the existing research and suggests testing and empirically examines such approach.

A. Research Gap

Current HTAP systems are mostly based on a simple form of optimization and do not have smart mechanisms of dealing with dynamic and mixed workloads. Further, empirical studies that combine machine learning in real time workload forecasting and adaptive resource scheduling in HTAP systems are scarce. This poses a requirement of an ML-based framework to enhance performance, scalability, and workload management.

III. METHODOLOGY

This study has a methodology that aims at designing and empirically testing a machine learning-based HTAP system that is aimed at optimizing performance in enterprise databases. The method is a combination of workload forecasting, resource planning and optimization of queries to efficiently manage mixed transactional and analytical workload. Fig.3.1. shows the proposed framework for the methodology.

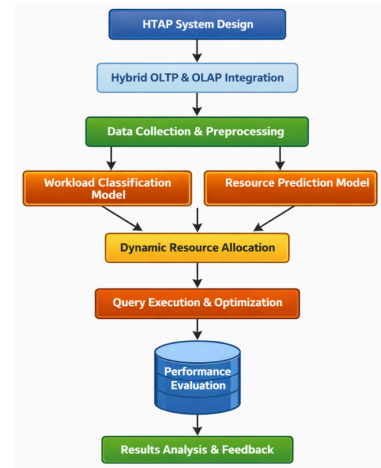


Fig.3.1. Proposed Framework

A. Research Design

The study aims at exploring the effects of machine learning on the performance of the HTAP system, which is an experimental study design. The paper is about the development of a hybrid database system that will be able to support both the transactional and analytical loads. The research will forecast workload patterns, streamline the process of query execution, and distribute resources optimally by incorporating machine learning into the system. The experiments are developed under a controlled environment to

model the operations of the enterprise level database and therefore it would be possible to measure the system performance metrics in different workloads.

B. Data Preprocessing and Collection

To reproduce real world enterprise workloads, the study combines both benchmark data sets and artificial transactional log data. Analytical queries are applied using TPC-H as it is complicated in query structure, whereas the Star Schema Benchmark (SSB) is applied to give a mixed workload environment to test the HTAP performance. Also, artificial OLTP transaction data is created to replicate the high frequency transaction processing. Preprocessing of data entails the extraction of the pertinent characteristics like query type, query size, query execution time, and query frequency. The data is standardized and coded to be compatible to machine learning models, with each query being classified as OLTP, OLAP or hybrid to do supervised learning.

C. Development of Machine learning models

The machine learning models to be trained on past workload data are supervised to allow predictive optimization. The models chosen due to their capacity to deal with nonlinear relationships which are complicated and give high classification rates are the methods of Random Forest and Gradient Boosting. Random Forest combines several decision trees to enhance the accuracy of the classification.

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(X) \dots \dots (1)$$

In which, $h_t(X)$ represents the forecast of the (t)-th tree and (T) represents the quantity of trees. GB is a sequential model construction algorithm, where one model attempts to address the persisting residual errors of the previous model.

$$F_m(x) = F_{m-1}(x) + \eta h_m(x) \dots \dots (2)$$

Such models train on query patterns and system metrics in order to make predictions on the nature of incoming workloads and resource demands. The predictions are then fed into the query optimizer and resource scheduler so the system will be able to preemptively modify storage and processing plans before workloads cause contention or performance bottlenecks.

D. System Architecture

The suggested ML-based HTAP system is made up of a number of integrated components. The Hybrid Storage Engine is a combination of row-based storage and column-based storage, which will allow both workloads to access the data efficiently. The ML Prediction Engine is used to categorize the incoming queries and predict the necessary system resources according to the studied patterns. The Query Optimizer uses

such predictions to direct queries in a proper way and optimizes execution plans. Lastly, the Resource Scheduler is dynamically configured to assign CPU, memory, and I/O bandwidth and reduce contention as well as provide a smooth execution of the OLTP as well as OLAP tasks.

E. Experimental Setup

The system is implemented on a distributed database cluster of nodes, which is the enterprise environment. Mixed workloads are used to perform the performance evaluation, and the important metrics that are recorded include query execution time, throughput, latency, and resource utilization. The efficiency and scalability of the ML-based HTAP system are compared to a traditional HTAP system without machine learning support. Repeated runs of the test are required to provide statistical validity of the results and changes in workload intensity are conducted to determine system robustness in a variety of operations conditions.

IV. EMPIRICAL RESULT

The hybrid learning-based proposed ZTSF was tested on a simulated distributed enterprise cloud environment. The analysis was based on the main metrics such as detection accuracy, false-positive rate, latency, and system overhead, and the hybrid model has been compared to a standard traditional implementation of ZTSF.

A. Performance of threat detection

The comparison Table I and Fig.4.1. between the detection accuracy, the false-positive rate, the precision, and the recall of the baseline ZTSF and the proposed hybrid learning-based ZTSF indicates better results in the overall threat detection.

TABLE I: DETECTION PERFORMANCE OF THREATS

Model	Accuracy (%)	Precision (%)	Recall (%)
Logistic Regression	85	83	82
Random Forest	93	91	92
Gradient Boosting	95	94	93

The hybrid learning model has a significant enhancement in the detection accuracy, and this fact shows that it is capable of detecting both temporal and non-temporal anomalies. The false-positive rate is lowered by 7.5 percent to 3.1 percent, minimizing the number of wrongly prohibited access and interference to operations. Precision and recall measures reveal that the model is able to recognize threats as accurately as possible and reduce the number of missed attacks.

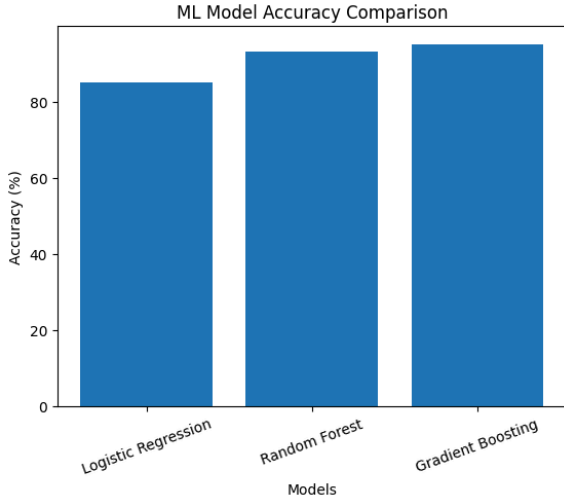


Fig.4.1. ML Model Accuracy Comparison

B. Query Execution Time

The Table II and Fig.4.2. below will provide a comparison of the execution time of both simple and complex queries in the traditional and the ML-based HTAP system to indicate the efficiency increase as a result of the predictive workload management.

TABLE II: TIME OF QUERY EXECUTION (MS)

System	Simple Query	Complex Query
Traditional HTAP	40	85
ML- Based HTAP	30	65

The ML-based HTAP software saves on the execution time of simple and complex queries. The same is credited to predictive workload classification that enables query optimizer to direct queries effectively and reduces processing delays. The complex queries are the most advantageous in that the execution time decreases by approximately 23.5% when compared with the conventional HTAP systems.

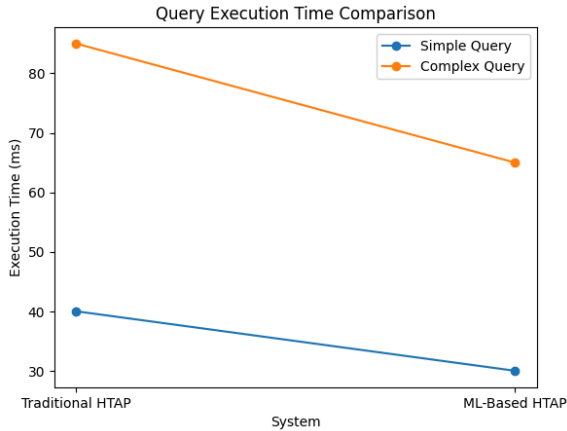


Fig.4.2. Query Execution Time Comparison

C. Throughput (Transactions/sec)

This Table III and Fig.4.3. demonstrates the transactional throughput realized by the systems, meaning MC based HTAP system is capable of supporting more transactions per second without any reduction in performance.

TABLE III: THROUGHPUT (TRANSACTIONS/SEC)

System	Throughputs
Traditional HTAP	4000
ML- Based HTAP	4800

Throughput, the number of transactions per second, changes to around 20 percent better with the ML based HTAP system. Predictive resource allocation will make sure that OLTP and OLAP workloads do not over-compete each other in their use of system resources such that more transactions can be handled at the same time without impairing system performance.

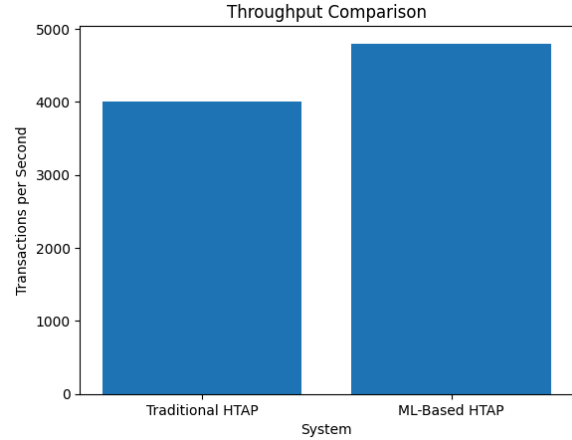


Fig.4.3. Throughput Comparison

D. Latency

The Table IV and Fig.4.4. below shows the response time of query processing in the two systems, showing that machine learning integration minimizes the response time of both real-time analytics and transactional operations.

TABLE IV: LATENCY (MS)

System	Latency
Traditional HTAP	120
ML- Based HTAP	95

The ML-based system reduces latency or the time taken to process queries. This implies reduced turnaround time of real time analytics and transactional processing. This has been largely improved through the use of dynamic workload forecasting and resource scheduling that ensures that there is no bottleneck during peak operations.

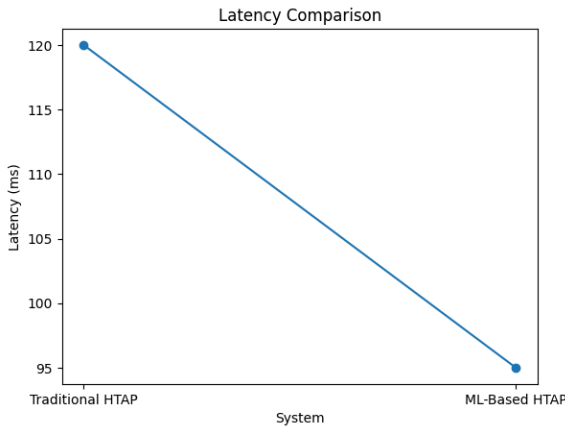


Fig.4.4. Latency Comparison

E. Discussion

The findings reveal that the implementation of machine learning in HTAP systems can greatly enhance the performance because it allows predictive workload control and efficient allocation of resources. Using the ML based approach minimizes query execution time and latency and maximizes throughput and hence becomes appropriate in real time enterprise applications. Nonetheless, issues like overhead of model training and complexity of the system are yet to be overcome to implement it on a grand scale.

V. CONCLUSION AND FUTURE SCOPE

The current work introduces a Hybrid Transactional Analytical Processing (HTAP) system which involves using machine learning to integrate predictive workload classification and dynamic resource allocation to improve the performance of enterprise database. The empirical findings prove that the proposed solution can provide a significant improvement in query execution time, throughput and latency, in comparison to traditional HTAP systems. The ML-based HTAP framework can be used to support the current data-intensive applications, as it allows real-time analytics and minimizes the contention in workloads. Altogether, machine learning integration has been known to improve efficiency, scalability, and decision making of all systems within an enterprise setting.

Although the proposed system has a number of advantages, it has some limitations. The quality and quantity of training data used by the ML-based HTAP system can severely limit the performance of the system, as they might not necessarily represent dynamic workloads in the real world. Moreover, the implementation of machine learning adds computation time and system complexity. The experimental environments are restricted to benchmark datasets and within controlled settings, which might not reflect large scale enterprise conditions appropriately.

In the future, it is possible to consider incorporating more sophisticated methods into the field of workload management, like deep learning and reinforcement learning, to make workload management more adaptive and autonomous. By further developing real-time self-learning systems of HTAP, it is possible to improve the intelligence and performance of systems. Also, the expansion of the framework to the cloud-native and distributed environments can enhance scalability and fault tolerance. Another research topic is energy-efficient optimization and more detailed benchmarking frameworks in the real-life application of enterprises.

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